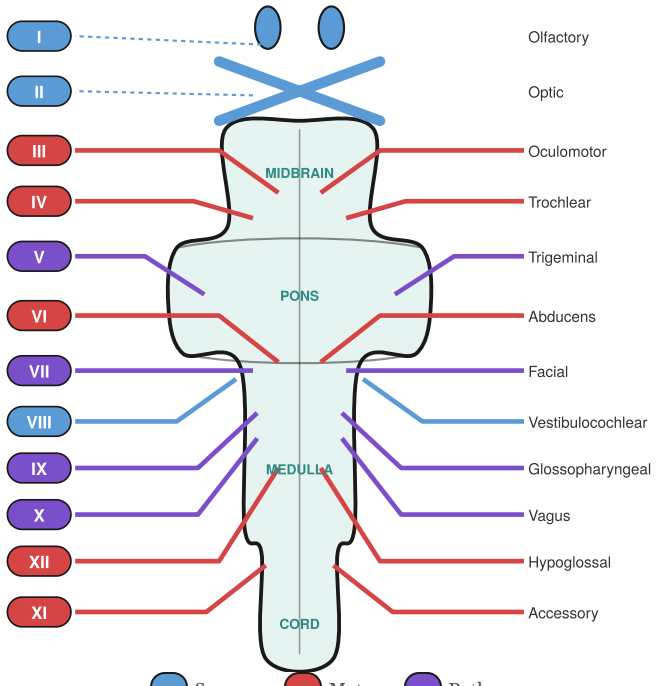
Harvard

Fundamentals of Neuroscience · MicroBachelors® Program

Cranial Nerves I–XII

MCB80x · name, type, exit level, function, and exam facts

VENTRAL VIEW — WHERE EACH NERVE EXITS



Schematic; nerves are paired (left and right).

Mnemonics

- **Names:** *Ooh, Ooh, Ooh, To Touch And Feel Very Good Velvet, Ah Heaven.*
- **Type:** *Some Say Marry Money, But My Brother Says Big Brains Matter More.* (S = sensory, M = motor, B = both)
- **Eye muscles:** LR_6SO_4, all the rest by III.

THE 12 CRANIAL NERVES

#	Nerve · exit	Type	Function · exam fact
I	**Olfactory** Forebrain	Sensory	Smell. **FACT** Only sense that skips the thalamus on the way to cortex.
II	**Optic** Forebrain	Sensory	Vision. **FACT** Nasal fibers cross at the optic chiasm.
III	**Oculomotor** Midbrain	Motor	Most eye movements, eyelid lift, pupil constriction. **FACT** Damage: eye “down and out,” dilated pupil.
IV	**Trochlear** Midbrain	Motor	Superior oblique: moves eye down and in. **FACT** Only nerve exiting the back of the brainstem.
V	**Trigeminal** Pons	Both	Face sensation; chewing muscles. **FACT** Trigeminal neuralgia: stabbing facial pain.
VI	**Abducens** Pons–medulla junction	Motor	Lateral rectus: moves eye outward. **FACT** Damage: eye can’t look to the side.
VII	**Facial** Pons–medulla junction	Both	Facial expression; taste (front ⅔ of tongue); tears, saliva. **FACT** Bell’s palsy: one side of face droops.
VIII	**Vestibulocochlear** Pons–medulla junction	Sensory	Hearing and balance. **FACT** Acoustic neuroma → hearing loss, vertigo.
IX	**Glossopharyngeal** Medulla	Both	Taste (back ⅓); swallowing; carotid body sensing. **FACT** Gag reflex: IX sensory, X motor.
X	**Vagus** Medulla	Both	Parasympathetic to heart, lungs, gut; voice box. **FACT** “Wanders” all the way to the abdomen.
XI	**Accessory** Medulla + cervical cord	Motor	Trapezius and sternocleidomastoid: shrug, turn head. **FACT** Roots rise from the upper spinal cord.
XII	**Hypoglossal** Medulla	Motor	Tongue movement. **FACT** Damage: tongue deviates toward the injured side.

Personal MCB80x study chart — not an official HarvardX/edX publication.